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Impacts of increasing volume of digital forensic data: A survey and future research challenges

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Abstract

A major challenge to digital forensic analysis is the ongoing growth in the volume of data seized and presented for analysis. This is a result of the continuing development of storage technology, including increased storage capacity in consumer devices and cloud storage services, and an increase in the number of devices seized per case. Consequently, this has led to increasing backlogs of evidence awaiting analysis, often many months to years, affecting even the largest digital forensic laboratories. Over the preceding years, there has been a variety of research undertaken in relation to the volume challenge. Solutions posed range from data mining, data reduction, increased processing power, distributed processing, artificial intelligence, and other innovative methods. This paper surveys the published research and the proposed solutions. It is concluded that there remains a need for further research with a focus on real world applicability of a method or methods to address the digital forensic data volume challenge.

Introduction

The increase in the number and volume of digital devices seized and lodged with digital forensic laboratories for analysis has been an issue raised over many years. This growth has contributed to lengthy backlogs of work (Gogolin, 2010, Parsonage, 2009). A significant growth in the size of storage media combined with the popularity of digital devices and the decrease in the price of these devices and storage media has led to a major issue affecting the timely process of justice. There is a growing volume of data seized and presented for analysis, often now consisting of many terabytes of data for individual investigations. This has resulted from;

(a) An increase in the number of devices seized per case.

(b) The number of cases with digital evidence is increasing (anecdotal information indicates the last case observed without digital evidence was at least 3 years old).

(c) The size of data on each individual item is increasing.

The increasing number of cases and devices seized is further compounded with the growing size of storage devices (Garfinkel, 2010). Existing forensic software solutions have evolved from the first generation of tools and are now beginning to address scalability issues. However, a gap remains in relation to analysis of large and disparate datasets. Every year the volume of data is increasing faster than the capability of processors and forensic tools can manage (Roussev et al., 2013).

Processing times are increasing with the increase in the amount of data required to be analysed. In the last decade, there have been many calls for research to focus on the timely analysis of large datasets (Garfinkel, 2010, Richard and Roussev, 2006a, Wiles et al., 2007) including the application of data mining techniques to digital forensic data in an endeavour to address the issue of the growing volume of information (Beebe and Clark, 2005, Palmer, 2001).

Serious implications relating to increasing backlogs include; reduced sentences for convicted defendants due to the length of time waiting for results of digital forensic analysis, suspects committing suicide whilst waiting for analysis, and suspects denied access to family and children whilst waiting for analysis (Shaw and Browne, 2013). In addition, employment can be affected for suspects under investigation for lengthy periods of time, and ongoing difficulties can be experienced by suspects and innocent persons when computers and other devices are seized, for example; the child of a suspect may have school assignments saved on a seized computer, or the partner of a suspect may have all their taxation or business information saved on a laptop.

In this paper we study literature examining the digital forensic data volume issue, including the volume of data, the growth of media, and research challenges. We review publications focussing on data mining, data reduction, triage, intelligence analysis, and other proposed methodologies. We then summarise the findings, and future directions for research are outlined in the conclusion.

We located material published in the last 15 years (i.e. 1/1/1999–14/6/2014) by searching various academic databases, including IEEE Xplore, ACM Digital Library, Google Scholar, and ScienceDirect using keywords such as; “Digital Forensic Data Volume”, “Computer Forensic Volume Problem”, “Forensic Data Mining”, “Digital Forensic Triage”, “Forensic Data Reduction”, “Digital Intelligence”, “Digital Forensic Growth”, and “Digital Forensic Challenges”. In addition, we browsed all papers published in Digital Investigation: The International Journal of Digital Forensics & Incident

Response, and The Journal of Digital Forensics, Security and Law. A summary table of key papers and topics is listed in Table 4 (see Discussion section).

Section snippets

Volume of data (1999–2009)

Digital forensics plays a crucial role in society across justice, security and privacy (Casey, 2014). Concerns regarding the increasing volume of data to be analysed in a digital forensic examinations have been raised for many years. McKemmish (1999) stated that the rapid increase in the size of storage media is probably the single greatest challenge to digital forensic analysis. In 2001, Palmer published the results of the first Digital Forensic Research Workshop (DFRWS), which included a …

Growth of media

In Survey section we outlined the many digital forensic papers discuss the problem of increasing volumes of data and devices, in this section we will focus on the growth of media. Moore's Law is the observation of an average doubling of the number of transistors on an integrated circuit every 18–24 months, which assists in predicting development of computer technology (Wiles et al., 2007). Kryder (as quoted by Walter, 2005) made the observation that in the space of 15 years, the storage density …

Data mining

Spafford (as cited in Palmer, 2001) listed data mining as a field of specialty which may assist in digital forensic analysis. Beebe (2009) also stated that the use of data mining techniques may be another solution to the volume challenge, and also that data mining has the potential to locate trends and information that may otherwise be undetected by human observation. Beebe also raised a list of topics for further research, including;

- a method for implementing subset collection, …
- how data mining …

…

Discussion

As outlined, there has been much discussion regarding the data volume challenge, and many calls for research into the application of data mining and other techniques to address the problem. Nevertheless, there has been very little published work in relation to a method or framework to apply data mining techniques, or other methods, to reduce or analyse the large volume of real-world data. In addition, the value of extracting or using intelligence from digital forensic data has had minimal …

Conclusion

Our literature survey identified that research gaps still remain in relation to the digital forensic data volume challenge. For example, there remains a need for research to be undertaken into data reduction techniques, data mining, intelligence analysis, and the use of open and closed source information. This should include research into the application of processes in a real world environment, and its acceptance in Courts and other tribunals.

Data Mining offers a potential solution to …

Acknowledgements

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Recommended articles

References (119)

R. Al-Zaidy *et al.*

Mining criminal networks from unstructured text documents

Digit Investig (2012)

W. Alink *et al.*

XIRAF – XML-based indexing and querying for digital forensics

Digit Investig (2006)

D. Ayers

[A second generation computer forensic analysis system](#)
Digit Investig (2009)

R.A.F. Bhoedjang *et al.*
Engineering an online computer forensic service
Digit Investig (2012)

C. Boyd
Time and date issues in forensic computing – a case study
Digit Investig (2004)

F. Breitinger *et al.*
On the database lookup problem of approximate matching
Digit Investig (2014)

F. Breitinger *et al.*
Automated evaluation of approximate matching algorithms on real data
Digit Investig (2014)

F. Buchholz *et al.*
A brief study of time
Digit Investig (2007)

A. Case *et al.*
Automated digital evidence discovery and correlation
Digit Investig (2008)

E. Casey
"Dawn raids" bring a new form in incident response
Digit Investig (2009)

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Cited by (198)

Multimedia big data computing and Internet of Things applications: A taxonomy and process model
2018, Journal of Network and Computer Applications

Show abstract

Future challenges for smart cities: Cyber-security and digital forensics

2017, Digital Investigation

> *Citation Excerpt :*
> …The threats to the individual components of the smart city do indeed affect the security of not only the smart city devices and software, but also the data that are to be stored in the Cloud. Consequently, uncertainty is created over data accuracy and its admissibility as forensic evidence (Quick and Choo, 2014). We now describe the threats posed to smart city data which are transmitted and stored in the Cloud:…

Show abstract

Copy-move forgery detection: Survey, challenges and future directions

2016, Journal of Network and Computer Applications

Show abstract

Efficient and Privacy-Preserving Outsourced Calculation of Rational Numbers ↗

2018, IEEE Transactions on Dependable and Secure Computing

Ensemble-based multi-filter feature selection method for DDoS detection in cloud computing ↗

2016, Eurasip Journal on Wireless Communications and Networking

A Survey on the Internet of Things (IoT) Forensics: Challenges, Approaches, and Open Issues ↗

2020, IEEE Communications Surveys and Tutorials

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